

Point

From Number Theory to Algebraic Geometry and Logic

Session 1 — Rings, Ideals, and Spectra

Complete lecture notes with proofs, examples, and exercises

Abstract

This session builds the first and most concrete incarnation of the course’s unifying concept: *the prime ideal as a point*. We develop the working language (rings, ideals, quotients, localization), prove the characterizations of prime and maximal ideals, establish Krull’s theorem from Zorn’s lemma, and then construct the *spectrum* $\text{Spec}(R)$ with its Zariski topology, showing that its points are precisely the prime ideals, its closed points the maximal ideals, and that an irreducible space has a *generic point* — “the whole space seen as one point”. Functoriality of Spec closes the session and previews the geometric acts.

Session plan (150 minutes)

Time	Block
0:00–0:10	Orientation: where this session sits in the course; the “point” theme
0:10–0:30	§1 Rings, ideals, quotients; correspondence theorem
0:30–0:45	§2 Localization; primes of $S^{-1}R$; local rings $R_{\mathfrak{p}}$
0:45–1:10	§3 Prime/maximal characterizations; nilradical = $\bigcap$ primes
1:10–1:20	<i>Break</i>
1:20–1:35	§4 Zorn’s lemma; Krull’s theorem
1:35–2:05	§5 The spectrum: Zariski topology, $D(f)$, quasi-compactness, generic points
2:05–2:15	§6 Functoriality of Spec
2:15–2:30	§7 Gallery of spectra; why Zariski is not Hausdorff; wrap-up

Conventions

Throughout, *ring* means commutative ring with multiplicative identity $1 \neq 0$; ring homomorphisms send 1 to 1; ideals are two-sided (automatic here); R denotes a ring and $I, J, \mathfrak{a}$ its ideals. We write $\subseteq$ for inclusion and $\subsetneq$ for proper inclusion.

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1 Rings, Ideals, and Quotients: the Working Language

Definition 1.1 (Ring). A **ring** is a set R with operations $+$ and $\cdot$ such that $(R, +)$ is an abelian group with identity 0 , multiplication is associative and commutative with identity $1 \neq 0$, and multiplication distributes over addition.

Example 1.2 (Running examples). • $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$; finite fields $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

- Polynomial rings $k[x], k[x, y]; \mathbb{Z}[x]$.
- Quadratic integer rings $\mathbb{Z}[i], \mathbb{Z}[\sqrt{-5}]$.
- Quotient rings $\mathbb{Z}/n\mathbb{Z}, k[x]/(f)$ (defined below).

These examples will carry every definition of the course; test each new notion on all of them.

Definition 1.3 (Ideal; generated; principal). An **ideal** $I \subseteq R$ is an additive subgroup with $rI \subseteq I$ for all $r \in R$. For $S \subseteq R$, the **ideal generated by** S is $(S) = \{\sum_{\text{finite}} r_i s_i : r_i \in R, s_i \in S\}$, the smallest ideal containing S . An ideal is **principal** if $I = (a)$ for one element; **proper** if $I \neq R$.

Definition 1.4 (Quotient ring). For an ideal I , the **quotient ring** R/I is the set of cosets $r + I$ with $(r + I) + (s + I) = (r + s) + I$ and $(r + I)(s + I) = rs + I$; the projection $\pi : R \rightarrow R/I, r \mapsto r + I$, is a surjective homomorphism with kernel I .

Proposition 1.5 (Universal property of the quotient). If $\varphi : R \rightarrow S$ is a homomorphism with $I \subseteq \ker \varphi$, there is a unique homomorphism $\bar{\varphi} : R/I \rightarrow S$ with $\bar{\varphi} \circ \pi = \varphi$.

Theorem 1.6 (Correspondence theorem). Let $I \subseteq R$ be an ideal. The maps

$$J \mapsto \pi^{-1}(J) \quad \text{and} \quad \mathfrak{a} \mapsto \pi(\mathfrak{a}) = \mathfrak{a}/I$$

give an inclusion-preserving bijection between ideals of R/I and ideals of R containing I . Under this bijection, prime ideals correspond to prime ideals and maximal to maximal.

Proof. Standard verifications show $\pi^{-1}(J)$ is an ideal of R containing I , and $\pi(\mathfrak{a})$ an ideal of R/I ; the two maps are inverse since π is surjective. For primality: if $\bar{x}\bar{y} \in \mathfrak{a}/I$ with lifts x, y , then $xy \in \mathfrak{a}$, so $x \in \mathfrak{a}$ or $y \in \mathfrak{a}$, hence $\bar{x} \in \mathfrak{a}/I$ or $\bar{y} \in \mathfrak{a}/I$. Maximality follows because $(R/I)/(\mathfrak{a}/I) \cong R/\mathfrak{a}$ (third isomorphism), and a ring is a field exactly when its only ideals are 0 and itself (Theorem 3.3 below). $\square$

Example 1.7. • Ideals of $\mathbb{Z}/n\mathbb{Z}$ correspond to divisors $d \mid n$; e.g. ideals of $\mathbb{Z}/12\mathbb{Z}$ are $(\bar{1}), (\bar{2}), (\bar{3}), (\bar{4}), (\bar{6}), (\bar{0})$; the maximal ones are $(\bar{2}), (\bar{3})$.

- $k[x]/(f)$: if f is irreducible this is a field (a finite extension of k); if $f = \prod f_i^{e_i}$ the ring has nilpotents and several maximal ideals $(\bar{f}_i)$.
- $\mathbb{C}[x, y]/(y - x^2) \cong \mathbb{C}[t]$: the coordinate ring of a parabola.

2 Localization

Definition 2.1 (Multiplicatively closed set; localization). A subset $S \subseteq R$ is **multiplicatively closed** if $1 \in S$ and $s, t \in S \Rightarrow st \in S$. The **localization** $S^{-1}R$ is the set of fractions r/s modulo $\frac{r}{s} = \frac{r'}{s'} \iff \exists t \in S : t(rs' - r's) = 0$, with the usual fraction operations; $\iota : R \rightarrow S^{-1}R, r \mapsto r/1$, is a homomorphism with $\ker \iota = \{r : \exists s \in S, sr = 0\}$.

Example 2.2. • $S = R \setminus \{0\}$, R a domain: $S^{-1}R = \text{Frac}(R)$; e.g. $\text{Frac}(\mathbb{Z}) = \mathbb{Q}$, $\text{Frac}(k[x]) = k(x)$.

- $S = R \setminus \mathfrak{p}$ for a prime $\mathfrak{p}$: write $R_{\mathfrak{p}}$; e.g. $\mathbb{Z}_{(p)} = \{a/b : p \nmid b\}$.
- $S = \{1, f, f^2, \dots\}$: write R_f ; e.g. $k[x]_x = k[x, x^{-1}]$.

Proposition 2.3 (Universal property of localization). *If $\varphi : R \rightarrow T$ sends every $s \in S$ to a unit of T , there is a unique $\tilde{\varphi} : S^{-1}R \rightarrow T$ with $\tilde{\varphi} \circ \iota = \varphi$.*

Proposition 2.4 (Primes of a localization). *The maps $\mathfrak{q} \mapsto \iota^{-1}(\mathfrak{q})$ and $\mathfrak{p} \mapsto \mathfrak{p}S^{-1}R$ give an inclusion-preserving bijection*

$$\mathrm{Spec}(S^{-1}R) \longleftrightarrow \{\mathfrak{p} \in \mathrm{Spec} R : \mathfrak{p} \cap S = \emptyset\}.$$

Proof. If $\mathfrak{q} \in \mathrm{Spec}(S^{-1}R)$ then $\mathfrak{p} := \iota^{-1}(\mathfrak{q})$ is prime and $\mathfrak{p} \cap S = \emptyset$ (elements of S map to units, and a unit cannot lie in a proper ideal). Conversely, for $\mathfrak{p}$ with $\mathfrak{p} \cap S = \emptyset$, the set $\mathfrak{p}S^{-1}R = \{p/s : p \in \mathfrak{p}, s \in S\}$ is an ideal; it is proper since $1 \notin \mathfrak{p}S^{-1}R$ (otherwise $t \cdot 1 \in \mathfrak{p}$ for some $t \in S$). It is prime: if $(r/s)(r'/s') = rr''/ss' \in \mathfrak{p}S^{-1}R$ then $trr' \in \mathfrak{p}$ for some $t \in S$, so $rr' \in \mathfrak{p}$ (as $t \notin \mathfrak{p}$), hence $r \in \mathfrak{p}$ or $r' \in \mathfrak{p}$. The two maps are mutually inverse by direct computation. $\square$

Corollary 2.5. *For a prime $\mathfrak{p}$, the ring $R_{\mathfrak{p}}$ is **local**: its only maximal ideal is $\mathfrak{p}R_{\mathfrak{p}}$, and $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \cong \mathrm{Frac}(R/\mathfrak{p}) =: \kappa(\mathfrak{p})$, the **residue field** at $\mathfrak{p}$.*

Proof. By Proposition 2.4, ideals of $R_{\mathfrak{p}}$ correspond to ideals of R contained in $\mathfrak{p}$; hence $\mathfrak{p}R_{\mathfrak{p}}$ is the unique maximal ideal. The residue field identification follows from $R_{\mathfrak{p}}/\mathfrak{p}R_{\mathfrak{p}} \cong (R/\mathfrak{p})_{\overline{0}\text{-complement}} = \mathrm{Frac}(R/\mathfrak{p})$. $\square$

3 Prime and Maximal Ideals

Definition 3.1 (Prime ideal). A proper ideal $\mathfrak{p} \subsetneq R$ is **prime** if $ab \in \mathfrak{p} \Rightarrow a \in \mathfrak{p}$ or $b \in \mathfrak{p}$.

Definition 3.2 (Maximal ideal). A proper ideal $\mathfrak{m} \subsetneq R$ is **maximal** if there is no ideal with $\mathfrak{m} \subsetneq I \subsetneq R$.

Theorem 3.3 (Characterizations). *For a proper ideal $I \subsetneq R$:*

- (a) I is prime $\iff R/I$ is an integral domain.
- (b) I is maximal $\iff R/I$ is a field.

Proof. (a) R/I a domain means $\bar{a}\bar{b} = 0 \Rightarrow \bar{a} = 0$ or $\bar{b} = 0$, i.e. $ab \in I \Rightarrow a \in I$ or $b \in I$.

(b) If R/I is a field and $I \subseteq J \subseteq R$, pick $x \in J \setminus I$ if $J \neq I$; then $\bar{x} \neq 0$ is invertible, so $1 = \bar{x}\bar{y} = \overline{xy}$, hence $1 \in J$, so $J = R$. Conversely if I is maximal and $\bar{x} \neq 0$, then $I + (x) = R$, so $1 = i + rx$ for some $i \in I$, whence $\bar{1} = \bar{r}\bar{x}$ and $\bar{x}$ is invertible; thus R/I is a field. $\square$

Corollary 3.4. *Every maximal ideal is prime.*

Proof. A field is an integral domain; apply Theorem 3.3. $\square$

Example 3.5. In $\mathbb{Z}$: (p) is maximal (and prime) for primes p ; (0) is prime but *not* maximal. In $k[x, y]$: (x) is prime (quotient $k[y]$, a domain) but not maximal; (x, y) is maximal (quotient k); (x^2) is neither (quotient has nilpotents).

Definition 3.6 (Nilradical). The **nilradical** $\mathrm{nil}(R) = \{x \in R : x^n = 0 \text{ for some } n \geq 1\}$.

Proposition 3.7. $\mathrm{nil}(R)$ is an ideal, and $\mathrm{nil}(R) = \bigcap_{\mathfrak{p} \in \mathrm{Spec} R} \mathfrak{p}$.

Proof. If $x^n = 0, y^m = 0$ then $(x + y)^{n+m} = 0$ by the binomial theorem (each term has x^i with $i \geq n$ or y^j with $j \geq m$), and $(rx)^n = 0$; so $\text{nil}(R)$ is an ideal. Every prime contains all nilpotents ($x^n \in \mathfrak{p} \Rightarrow x \in \mathfrak{p}$), giving $\text{nil}(R) \subseteq \bigcap \mathfrak{p}$.

For the reverse, suppose $x \notin \text{nil}(R)$; then $S = \{1, x, x^2, \dots\}$ is multiplicatively closed with $0 \notin S$. Let Σ be the set of ideals disjoint from S , ordered by inclusion; $\Sigma \neq \emptyset$ ($(0) \in \Sigma$). Zorn's lemma (Section 4) gives a maximal element $\mathfrak{p} \in \Sigma$. We claim $\mathfrak{p}$ is prime: if $a, b \notin \mathfrak{p}$ then $\mathfrak{p} + (a)$ and $\mathfrak{p} + (b)$ meet S , so $x^i = p_1 + ra, x^j = p_2 + sb$; multiplying, $x^{i+j} \in \mathfrak{p} + (ab)$, so $\mathfrak{p} + (ab)$ meets S , forcing $ab \notin \mathfrak{p}$. Hence $\mathfrak{p} \in \text{Spec } R, x \notin \mathfrak{p}$, so $x \notin \bigcap \mathfrak{p}$. $\square$

Definition 3.8 (Jacobson radical). The **Jacobson radical** $J(R) = \bigcap_{\mathfrak{m} \text{ maximal}} \mathfrak{m}$. Always $\text{nil}(R) \subseteq J(R)$.

Remark 3.9. $\text{nil}(R)$ measures “functions vanishing to infinite order everywhere”; $J(R)$ measures elements invertible-modulo-every-point. For $\mathbb{Z}$ both are 0; for $\mathbb{Z}/p^n\mathbb{Z}$ both equal $(\bar{p})$.

4 Zorn's Lemma and the Existence of Maximal Ideals

Theorem 4.1 (Zorn's lemma (axiom)). *If every chain in a nonempty partially ordered set $(\Sigma, \leq)$ has an upper bound in Σ , then Σ has a maximal element.*

Theorem 4.2 (Krull). *Every proper ideal of a ring R is contained in a maximal ideal. In particular every nonzero ring has at least one maximal ideal, and $\text{Spec } R \neq \emptyset$.*

Proof. Let $I \subsetneq R$ and let Σ be the set of proper ideals containing I , ordered by inclusion; $\Sigma \neq \emptyset$. If $\{I_\alpha\}$ is a chain, set $J = \bigcup_\alpha I_\alpha$: J is an ideal (unions of chains of ideals are ideals) and $J \neq R$ (else $1 \in I_\alpha$ for some α). So $J \in \Sigma$ is an upper bound; Zorn gives a maximal element $\mathfrak{m} \in \Sigma$, which is a maximal ideal containing I . Since (0) is proper, a maximal ideal exists; and maximal ideals are prime, so $\text{Spec } R \neq \emptyset$. $\square$

Remark 4.3 (Where the axiom reappears). The same argument (or its relatives) returns in Session 3 (existence of algebraic closures) and Session 58 (Deligne's theorem: coherent topoi have enough points). Flag it for students: these are the three places the course genuinely uses choice.

5 The Spectrum of a Ring

Definition 5.1 (Spectrum). The **spectrum** of R is the set $\text{Spec } R = \{\mathfrak{p} \subseteq R : \mathfrak{p} \text{ prime}\}$.

Definition 5.2 (Closed sets $V(I)$). For an ideal I , set $V(I) = \{\mathfrak{p} \in \text{Spec } R : I \subseteq \mathfrak{p}\}$. Declare the sets $V(I)$ to be the **closed sets** of the **Zariski topology** on $\text{Spec } R$.

Proposition 5.3 (Zariski axioms). (a) $V(0) = \text{Spec } R$ and $V(R) = \emptyset$.

(b) $V(\sum_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$.

(c) $V(I \cap J) = V(IJ) = V(I) \cup V(J)$.

Hence the $V(I)$ are exactly the closed sets of a topology.

Proof. (a) Every prime contains 0; no proper prime contains 1. (b) $\mathfrak{p} \supseteq \sum_\alpha I_\alpha \iff \mathfrak{p} \supseteq I_\alpha$ for all α . (c) $IJ \subseteq I \cap J \subseteq I$ gives $V(I) \subseteq V(I \cap J) \subseteq V(IJ)$; for the reverse, if $\mathfrak{p} \supseteq IJ$ but $\mathfrak{p} \not\supseteq I$ and $\mathfrak{p} \not\supseteq J$, pick $a \in I \setminus \mathfrak{p}, b \in J \setminus \mathfrak{p}$; then $ab \in IJ \subseteq \mathfrak{p}$ with $a, b \notin \mathfrak{p}$, contradicting primality. So $V(IJ) = V(I) \cup V(J)$, and the displayed inclusions force $V(I \cap J) = V(IJ)$ as well. $\square$

Lemma 5.4. $V(I) = V(\text{rad } I)$, where $\text{rad } I = \{x : x^n \in I\}$; moreover $\text{rad } I = \bigcap_{\mathfrak{p} \supseteq I} \mathfrak{p}$.

Proof. Apply Proposition 3.7 in R/I : $\text{nil}(R/I) = \text{rad } I/I = \bigcap_{\mathfrak{p} \supseteq I} \mathfrak{p}/I$. Since $V(I)$ depends only on the set of primes containing I , and $\text{rad } I$ has the same containing primes as I , the equality $V(I) = V(\text{rad } I)$ follows. $\square$

Definition 5.5 (Basic opens). For $f \in R$, $D(f) = \text{Spec } R \setminus V((f)) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$.

Proposition 5.6 (The $D(f)$ form a basis). (a) $D(f) \cap D(g) = D(fg)$.

(b) $\bigcup_i D(f_i) = \text{Spec } R \iff (f_i : i) = R$.

(c) The $D(f)$ form a basis of the Zariski topology.

Proof. (a) $f \notin \mathfrak{p}$ and $g \notin \mathfrak{p} \iff fg \notin \mathfrak{p}$ (primality). (b) $\bigcup_i D(f_i) = \text{Spec } R$ means no prime contains all f_i , i.e. $V((f_i)) = \emptyset$; by Lemma 5.4 this means $\text{rad}((f_i)) = \bigcap_{\mathfrak{p}} \mathfrak{p}$ over primes containing (f_i) is all of R , i.e. $(f_i) = R$ (else Krull gives a maximal, hence prime, ideal containing it). (c) Every open is $\text{Spec } R \setminus V(I) = \bigcup_{f \in I} D(f)$, and (a) gives stability under finite intersection; hence the $D(f)$ form a basis. $\square$

Theorem 5.7 (Quasi-compactness of Spec). *Spec R is quasi-compact: every open cover has a finite subcover. In fact each $D(f)$ is quasi-compact.*

Proof. Let $D(f) \subseteq \bigcup_i D(g_i)$. Then $V((f)) \supseteq \bigcap_i V((g_i)) = V((g_i : i))$, so by Lemma 5.4, $f^n \in (g_i : i)$ for some n : $f^n = \sum_{j=1}^N r_j g_{i_j}$. Any prime containing all g_{i_j} contains f^n , hence f ; contrapositively $D(f) \subseteq \bigcup_{j=1}^N D(g_{i_j})$. Taking $f = 1$ gives quasi-compactness of $\text{Spec } R$. $\square$

Lemma 5.8 (Closure of a point). For $\mathfrak{p} \in \text{Spec } R$, the closure of $\{\mathfrak{p}\}$ is $V(\mathfrak{p}) = \{\mathfrak{q} : \mathfrak{q} \supseteq \mathfrak{p}\}$.

Proof. $\overline{\{\mathfrak{p}\}}$ is the smallest closed set containing $\mathfrak{p}$, i.e. the intersection of all $V(I)$ with $\mathfrak{p} \in V(I)$, i.e. all $V(I)$ with $I \subseteq \mathfrak{p}$; the smallest such closed set is $V(\mathfrak{p})$ since $I \subseteq \mathfrak{p} \Rightarrow V(\mathfrak{p}) \subseteq V(I)$. $\square$

Corollary 5.9 (Closed points). $\mathfrak{p}$ is a closed point of $\text{Spec } R \iff \mathfrak{p}$ is a maximal ideal.

Proof. $\{\mathfrak{p}\}$ closed $\iff V(\mathfrak{p}) = \{\mathfrak{p}\} \iff$ no prime strictly contains $\mathfrak{p} \iff \mathfrak{p}$ maximal. $\square$

Definition 5.10 (Generic point). A point $\eta \in \text{Spec } R$ is **generic** if $\overline{\{\eta\}} = \text{Spec } R$.

Example 5.11. If R is an integral domain then (0) is the unique generic point: $\overline{\{(0)\}} = V(0) = \text{Spec } R$. In $\text{Spec } \mathbb{Z}$ the generic point is (0) ; every nonempty open contains it. More generally the generic points of $\text{Spec } R$ are the minimal primes.

Definition 5.12 (Specialization). We say $\mathfrak{p}$ **specializes** to $\mathfrak{q}$ (write $\mathfrak{p} \rightsquigarrow \mathfrak{q}$) if $\mathfrak{q} \in \overline{\{\mathfrak{p}\}}$, i.e. $\mathfrak{p} \subseteq \mathfrak{q}$. Closed points are “the most special”; generic points “the most general”.

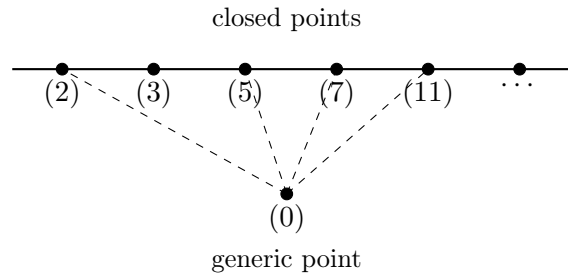


Figure 1: $\text{Spec } \mathbb{Z}$: one closed point per prime, plus the generic point (0) whose closure is everything.

6 Functoriality of the Spectrum

Proposition 6.1. *Let $\varphi : R \rightarrow S$ be a ring homomorphism. For $\mathfrak{q} \in \operatorname{Spec} S$, the preimage $\varphi^{-1}(\mathfrak{q})$ is a prime ideal of R ; the assignment*

$$\varphi^* : \operatorname{Spec} S \longrightarrow \operatorname{Spec} R, \quad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q})$$

is continuous. Moreover $(\psi \circ \varphi)^ = \varphi^* \circ \psi^*$ and $(\operatorname{id})^* = \operatorname{id}$, so Spec is a contravariant functor $\mathbf{Ring} \rightarrow \mathbf{Top}$.*

Proof. Primality: if $ab \in \varphi^{-1}(\mathfrak{q})$ then $\varphi(ab) = \varphi(a)\varphi(b) \in \mathfrak{q}$, so $\varphi(a) \in \mathfrak{q}$ or $\varphi(b) \in \mathfrak{q}$. Continuity: for an ideal $I \subseteq R$, $(\varphi^*)^{-1}(V(I)) = \{\mathfrak{q} : I \subseteq \varphi^{-1}(\mathfrak{q})\} = \{\mathfrak{q} : \varphi(I) \subseteq \mathfrak{q}\} = V(\varphi(I)S)$, a closed set. Functoriality is direct computation. $\square$

Example 6.2. • $\mathbb{Z} \hookrightarrow \mathbb{Q}$: $\operatorname{Spec} \mathbb{Q} = \{(0)\} \rightarrow \operatorname{Spec} \mathbb{Z}$ lands on the generic point (0) .

- $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/n\mathbb{Z}$: $\operatorname{Spec}(\mathbb{Z}/n\mathbb{Z}) \rightarrow \operatorname{Spec} \mathbb{Z}$ lands on the primes dividing n .
- $k \hookrightarrow k[x]$: the closed point $(x - a)$ maps to (0) — “a point of the line lies over the generic point of the ground field”; this is the first hint that fibres matter (S30).

Remark 6.3 (Why contravariant, and why it matters). Points *pull back*: a map of rings $R \rightarrow S$ lets an S -point be viewed as an R -point. This is exactly the functor-of-points philosophy made rigorous in Session 29 ($X \cong \operatorname{Hom}(-, X)$) and again in Session 54 (a topos point is a morphism $\mathbf{Set} \rightarrow \mathcal{E}$). The theme “point = morphism into the object” starts here.

7 A Gallery of Spectra; the Zariski Topology is Strange

Example 7.1 (Gallery). • $\operatorname{Spec} \mathbb{Z}$: closed points (p) , one generic (0) (figure above).

- $\operatorname{Spec} \mathbb{C}[x]$: closed points $(x - a)$, $a \in \mathbb{C}$, plus generic (0) — the affine line.
- $\operatorname{Spec} \mathbb{R}[x]$: closed points $(x - a)$ and $(x^2 + bx + c)$ with $b^2 - 4c < 0$ (conjugate pairs), plus (0) .
- $\operatorname{Spec} \mathbb{C}[x, y]$: closed points $(x - a, y - b)$; curve-generic points (f) for irreducible f ; the generic (0) . A two-dimensional space with three layers of points.
- $\operatorname{Spec} \mathbb{Z}_{(p)}$: exactly two points, $(pR_{(p)})$ closed and (0) generic (Corollary 2.5).
- $\operatorname{Spec} \mathbb{C}[x, y]/(xy)$: two crossing lines; the crossing point (x, y) is where the space fails to be a manifold — a first taste of singularities (S25, S30).

Remark 7.2 (Not Hausdorff, but T_0). In $\operatorname{Spec} \mathbb{Z}$, any two nonempty opens meet: $D(m) \cap D(n) = D(mn) \neq \emptyset$ since $(mn) \neq \mathbb{Z}$ has a prime over it. So the Zariski topology is never Hausdorff for infinite spectra. It is however T_0 : distinct points have distinct closures (Lemma 5.8). This “coarseness” is a feature: the topology remembers *algebraic* specialization, not metric proximity. Compare with the p -adic and analytic topologies of Sessions 17–19 and 37–38, and with the étale topology of Session 52, which repairs exactly this deficiency.

8 Exercises

Exercise 1. List all ideals of $\mathbb{Z}/36\mathbb{Z}$; identify the prime and the maximal ones; draw the inclusion poset.

Exercise 2. Prove that $k[x, y]/(y - x^2) \cong k[t]$. What happens for $k[x, y]/(y^2 - x^3)$? (Is it a domain? Is it isomorphic to $k[t]$?)

Exercise 3. Show $\text{nil}(\mathbb{Z}/n\mathbb{Z})$ equals the ideal generated by the product of the distinct primes dividing n . Compute $\text{nil}(\mathbb{Z}/72\mathbb{Z})$.

Exercise 4. Prove that a ring R is a field $\iff$ its only ideals are 0 and $R \iff$ every nonzero element is a unit. Deduce that a homomorphism from a field is injective or zero.

Exercise 5. Describe $\text{Spec } \mathbb{R}[x]$ completely (closed points, generic point) and compare with $\text{Spec } \mathbb{C}[x]$. Which points of $\text{Spec } \mathbb{C}[x]$ lie over a given point of $\text{Spec } \mathbb{R}[x]$ under the map of Proposition 6.1?

Exercise 6. Prove that $D(f) = \emptyset \iff f \in \text{nil}(R)$, and that $D(f) = \text{Spec } R \iff f$ is a unit.

Exercise 7. Show $\text{Spec } R$ is irreducible (not a union of two proper closed subsets) $\iff \text{nil}(R)$ is a prime ideal $\iff \text{Spec } R$ has a generic point.

Exercise 8. Let $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}[i]$ be the inclusion. Describe the fibres of $\varphi^* : \text{Spec } \mathbb{Z}[i] \rightarrow \text{Spec } \mathbb{Z}$ over (2), (3), (5), and (0). (This is decomposition of primes in a quadratic extension — the concrete preview of Sessions 9 and 20.)

Exercise 9. Prove that the localization functor $S^{-1}(-)$ is exact: it sends short exact sequences of R -modules to short exact sequences. (Uses Session 2 language; do after Session 2 if preferred.)

Exercise 10. Give an example of a ring whose Jacobson radical is strictly larger than its nil-radical, and one where they coincide but are nonzero.

Where This Session Feeds the Course

- **Depends on:** nothing (first session); P1–P3 supply group/topological language as needed.
- **Feeds:** S2 (modules over R , fibres $M \otimes_R \kappa(\mathfrak{p})$); S24 (Dedekind domains, prime factorization); S28 (schemes = glued spectra with structure sheaf); S30 (morphisms, fibres); S46 (decomposition/inertia over primes); S52–S54 (Zariski site, topos points); Zorn reused in S3, S58.
- **Unifying thread:** the prime ideal is the first incarnation of “point”; the generic point is “the whole space seen as one point”; specialization is the arrow from general to special; Spec functoriality is the first appearance of “point = morphism into the object”.

References for this session

- [1] Atiyah, M.F. & Macdonald, I.G. *Introduction to Commutative Algebra*, ch.1–3.
- [2] Eisenbud, D. *Commutative Algebra with a View Toward Algebraic Geometry*, ch.2, 4.
- [3] Vakil, R. *The Rising Sea*, ch.3–4.
- [4] Hartshorne, R. *Algebraic Geometry*, ch.II §2 (for the scheme-theoretic sequel).
- [5] Matsumura, H. *Commutative Ring Theory*, ch.1–2 (supplementary).

End of Session 1 — next: Session 2, Modules, Tensor Products, and Exact Sequences